Qn. Set Code-7(SOT)

Semester: 4th
Programme: B.Tech
Branch: CSE, IT, CSCE, CSSE, E&CSc

SPRING END SEMESTER EXAMINATION-2025

4th Semester B.Tech

DATABASE MANAGEMENT SYSTEMS

CS20006/ IT20006/ CM20006/ CC20006/ CS2004

(For 2023 & Previous Admitted Batches)

Time: 2 Hours 30 Minutes | Full Marks: 50

Answer any FIVE questions.
Question paper consists of two SECTIONS i.e. A and B.
Section A is compulsory.
Attempt any Four question from Sections B.
The figures in the margin indicate full marks.
Candidates are required to give their answers in their own words as far as practicable and all parts of a question should be answered at one place only.

SECTION-A

1. Answer the following questions. [1 × 10]

(a) A minimum cardinality of 0 specifies _____ type of participation.

(b) Let E1 and E2 be two entity sets in ER diagram with simple single-valued attributes; R1 and R2 are two relationships between E1 and E2, where R1 is one-to-many and R2 is many-to-many. R1 and R2 do not have any attributes of their own. What is the minimum number of tables required to represent this situation in the relational model?

(c) Consider the relations Stud(roll, name, address) and Enroll(roll, course_no, course_name). The numbers of tuples in Stud and Enroll tables are 120 and 8 respectively. What are the maximum and minimum numbers of tuples that can be presented in Stud NATURAL JOIN Enroll?

(d) Which normal form is considered adequate for normal relational database design? Justify your answer.

(e) Consider the relation R(E, F, G, H, I, J, K, L, M, N) and the set of functional dependencies {EF→G, F→IJ, EH→KL, K→M, L→N}. What is the candidate key for R?

(f) What is a view? How do views support logical data independence?

(g) Find the equivalent query in relational algebra for the below SQL query:

SELECT DISTINCT a1, a2, .. am from R1, R2, .. Rn where P;

(h) How is the log used in transaction rollback and crash recovery?

(i) Explain what a RAID system is and how it improves performance and reliability.

(j) Consider a B+ tree in which the maximum number of keys in a node is 5. What is the minimum number of keys in any non-root node?

SECTION-B

2. (a) Create a B – Tree with order 5 for the data members: [5]

10, 25, 34, 23, 56, 45, 76, 84, 41, 74, 17.

Also, draw the B – Tree after the deletion of data item 17 followed by deletion of data item 10.

(b) What do you mean by the integrity constraints? Explain each type with the suitable example. [5]

3. (a) Consider the following relations: [5]

Student(snum, sname, major, level, age)

Class(cname, time, room, fid)

Enrolled(snum, cname)

Faculty(fid, fname, deptid)

Write the following queries in relational algebra.

a. Find the names of all Junior level students who are enrolled in a class taught by 'Mihir'.

b. Find the names of all classes that either conducted in room 'A107' or have 5 or more students enrolled.

c. Find the names of all students who are enrolled in 2 classes that are conducted at the same time.

d. Find the level and the average age of students for that level, for each level.

e. Find the names of students not enrolled in any class.

(b) Explain the join and selection operators in a RDBMS. How can join operators can be used to simulate a selection? [5]

4. (a) Consider the relation Stud(Roll, Name, Course, Grade) with the following functional dependencies: [5]

Name, Course -> Grade,

Roll, Course-> Grade,

Name-> Roll,

Roll-> Name

What will be the highest normal form of this relation?

(b) Consider the relation R(A, B, C, D, E) with functional dependencies {AB→C, C→E, B→D, E→A}. If the relation R is decomposed to R1(B, C, D) and R2(A, C, E), check whether the decomposition is lossless or not. Also, check whether the decomposition is dependency preserving or not. [5]

5. (a) Consider the schedules S1 and S2 for the transactions T1, T2 and T3. [5]

S1: r1(x), r3(y), r3(x), r2(y), r2(z), w3(y), w2(z), r1(z), w1(x), w1(z)

S2: r1(x), r3(y), r2(y), r3(x), r1(z), r2(z), w3(y), w1(x), w2(z), w1(z)

Which of the schedule is conflict serializable?

(b) In timestamp-based concurrency control, transactions are assigned a timestamp at startup; how is it used to ensure serializability? Explain with suitable example. How does the Thomas Write Rule improve concurrency? [5]

6. (a) Draw an ER diagram for the airport database. The details are: [5]

Every airplane has a registration number, and each airplane is of a specific model.

The airport accommodates a number of airplane models, and each model is identified by a model number and has a capacity and a weight.

A number of technicians work at the airport. You need to store the name, SSN, address, phone number and salary of each technician.

Each technician is an expert on one or more plane model(s), and his or her expertise may overlap with that of other technicians. The information about technicians must also be recorded.

Traffic controllers must have an annual medical examination. For each traffic controller, you must store the date of the most recent exam.

All airport employees (including technicians) belong to a union. You must store the union membership number of each employee. You can assume that each employee is uniquely identified by a social security number (SSN).

The airport has a number of tests that are used periodically to ensure that airplanes are still airworthy. Each test has a Federal Aviation Administration (FAA) test number, a name and a maximum possible score.

The FAA requires the airport to keep track of each time a given airplane is tested by a given technician using a test. For each testing event, the information needed is the date, the number of hours the technician spent doing the test and the score the airplane received on the test.

Make necessary assumptions. Cleary specify the cardinalities and primary keys.

(b) Map the above ER diagram to create the relational model. Indicate primary keys as well as referential integrity constraints. [5]

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